



Technical Service Bulletin

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Belt Shredding

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- QSX15 CM570

Options:

- FA1542
- FA1657

Issue

Symptom:

- Progressive belt shredding caused by belt jump. See Figures 1 and 2 below.
- Belt shredding can lead to loss of drive belt which will cause high coolant temperatures.

Root Cause:

- High fan center allows excessive belt deflection between fan pulley and crankshaft pulley.



Figure 1, Belt Shred Shown Over Waterpump Pulley, Missing Two Leftmost Ribs.



Figure 2, Removed belt with belt shred symptom – top ribs are missing

Description of Change

A new 3-idler pulley bracket for engines with specific high mount fan centers (fan drive options FA1542 and FA1657). The 3-idler pulley bracket is mounted on fan bracket between fan and crankshaft pulley. See Figure 3 below.

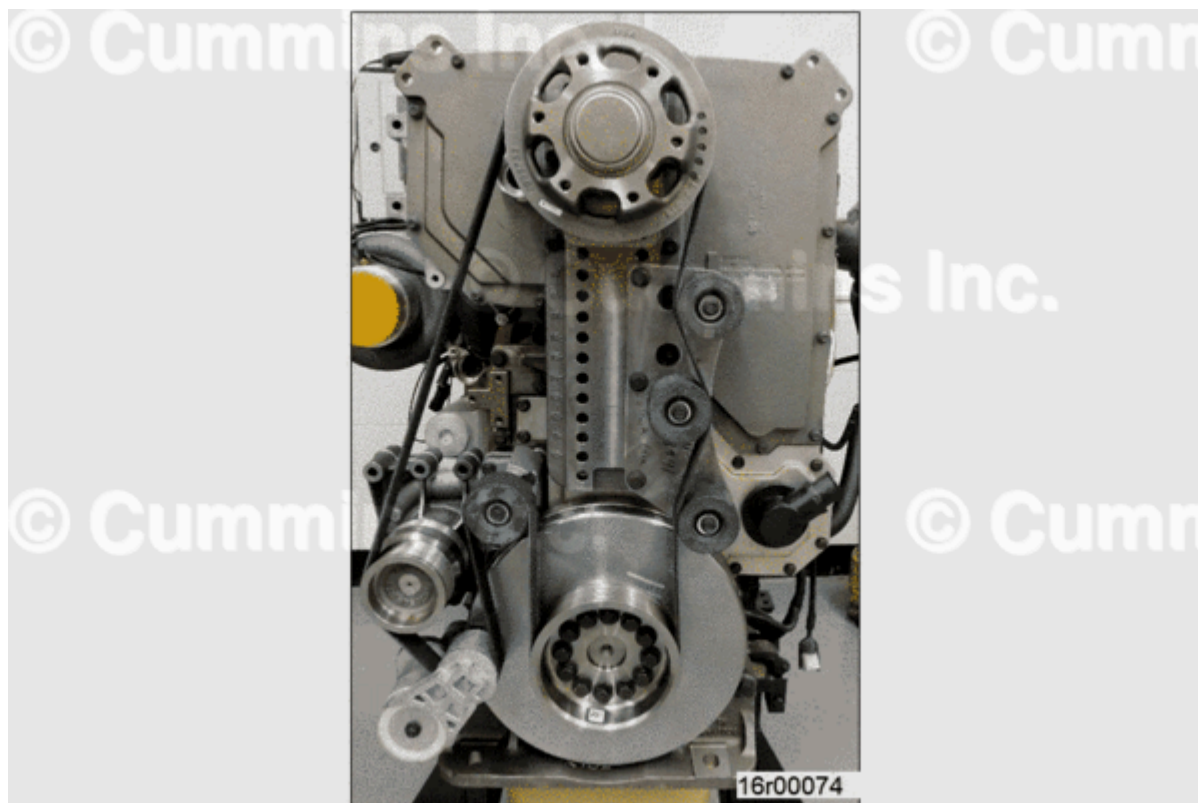


Figure 3. Updated drive layout with 3-idler bracket

Verification

Verify engine was built with FA1542 or FA1657.

Inspect belt for shredding or missing ribs on outside or inside of belt before removing from engine.

Compare belt to Figure 1 and Figure 2 above to confirm malfunction mode is belt shred.

Resolution

- If fan drive option and malfunction mode match verification, a 3-idler bracket is to be installed.
- Engines with FA1542 will need to have a larger cooling fan belt tensioner installed on low engine mount and an idler pulley added to center of drive.
- Engines with FA1657 will already have this cooling fan belt tensioner and idler pulley installed. The information below shows parts and instructions for installing new parts for FA1542 and FA1657 in separate sections. Verify which drive is installed on engine and reference correct instructions.
- Verify engines built with FA1542 have not been updated to FA1657 according to TSB120285 before proceeding.

Service Instructions

See QSX15 CM570 Service Manual, Bulletin number 3666239.

Installation Instructions for Engines with FA1542

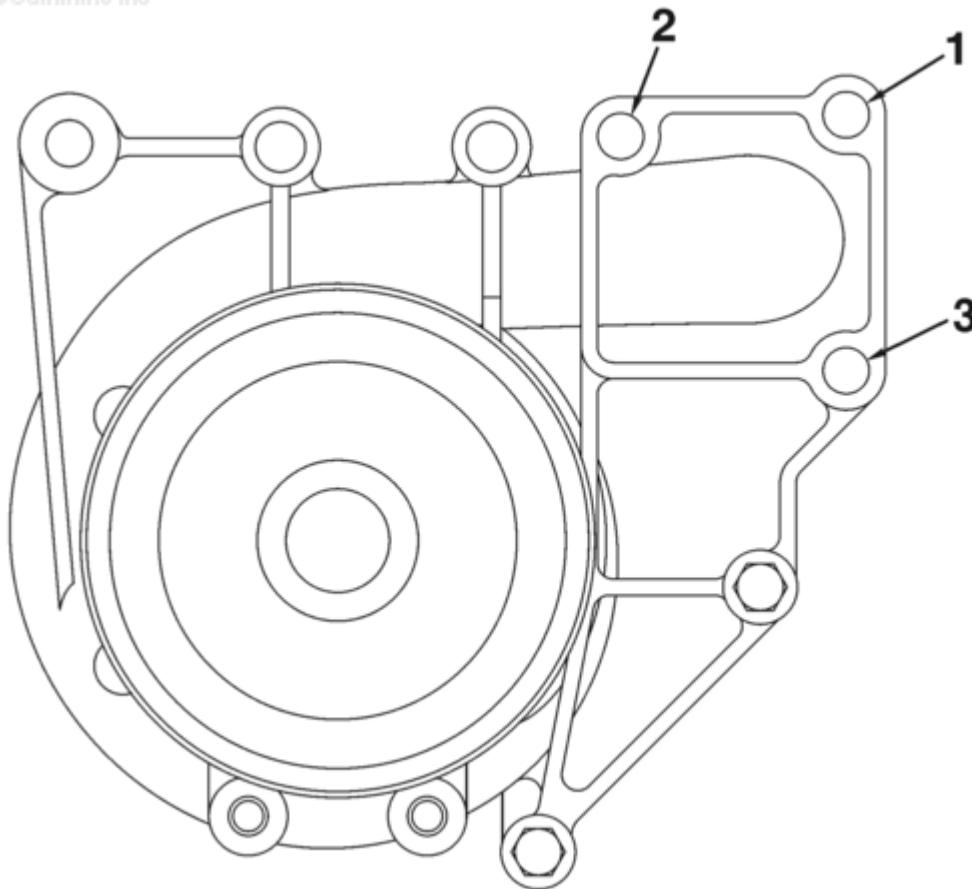
Verify that engine currently has FA1542 installed. See Figure 1 below. The belt cooling fan belt tensioner will be installed in middle of drive, directly above crankshaft pulley.



Figure 4, FA1542 Drive Layout.

Removal of FA1542 components:

- Remove fan drive belt, belt cooling fan belt tensioner, and cooling fan belt tensioner support from fan bracket.
 - See QSX15 CM570 Service Manual, Bulletin number 3666239. Reference Procedures 008-002 and 008-080 in Section 8.
 - Remove three capscrews on right side of water pump. See Figure 5 below.



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Figure 5, Water Pump Capscrews to be Removed

Installation of cooling fan belt tensioner and idler pulley:

- Install belt cooling fan belt tensioner bracket, Part Number 3681191. Use capscrews removed from water pump. Capscrew 1 is Part Number 3681390 and capscrews 2 and 3 are Part Number 3093936. Capscrew 1 requires washer Part Number 3036460 and Capscrew 3 requires washer Part Number 3009330.
 - Tighten all capscrews to 58 N-m [43 ft-lb]
- Install pilot adapter Part Number 3681587, using dowel pin Part Number 70760, and idler pulley Part Number 3681587.
 - Reference Procedure 008-062 in Section 8.
- Install low-mount large cooling fan belt tensioner Part Number 3104029.
 - Reference Procedure 008-080 in Section 8 – "For low-mount (large cooling fan belt tensioner)".

Installation of tight side idler bracket assembly and fan drive belt:

- Remove fan support bracket (ladder bracket).
 - Reference Procedure 008-088 in Section 8.
- Install idler pulley bracket Part Number 5575127 to fan support bracket. bottom hole of idler pulley bracket should line up with bottom right hole of fan support bracket. See Figure 3. Use hex nut Part Number 3035803 and capscrew Part Number 3093940.
 - Tighten capscrews to 75 N-m [55 ft-lb]
- Reinstall fan support bracket.
 - Reference Procedure 008-088 in Section 8.

- Install idler pulley Part Number 3681587 to idler pulley bracket Part Number 5575127. Use capscrew Part Number 3903112 and washer Part Number 3026269.
 - Tighten capscrew to 46 N-m [34 ft-lb]
- Install dust shield Part Number 3935013 to idler pulley by pressing on outside diameter of shield to prevent damage to dust shield.
 - Reference Procedure 008-030 in Section 8.
- Install fan drive belt, Part Number 3103848.
 - Reference Procedure 008-002 in Section 8.

Installation Instructions for Engines with FA1657

- Verify engine has FA1657 option. See Figure 6 below.
- Tensioner will be installed on low engine mount and there will be an idler pulley directly above crankshaft pulley. There will be a tight side idler pulley, too.

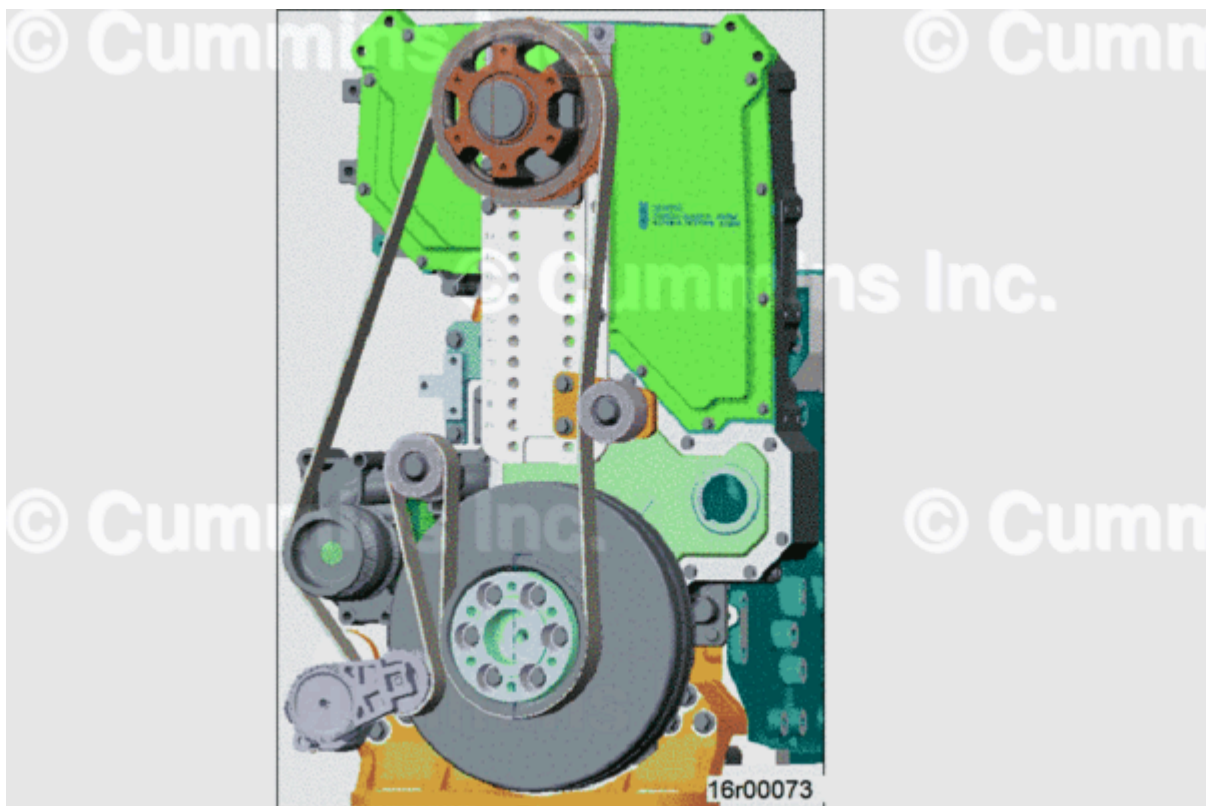


Figure 6, FA1657 Drive Layout.

Removal of FA1657 Components

- Remove fan drive belt.
 - Reference Procedure 008-002 in Section 8.
- Remove fan support bracket (ladder bracket).
 - Reference Procedure 008-088 in Section 8.
- Remove tight side idler bracket that is attached to fan support bracket with two capscrews.
 - Do **not** reinstall fan support bracket until new idler bracket is installed.

Installation of tight side idler bracket assembly and fan drive belt:

- Install idler pulley bracket Part Number 5575127 to fan support bracket (removed in previous step). The bottom hole of idler pulley bracket should line up with bottom right hole of fan support bracket. See Figure 3. Use hex nut Part Number 3035803 and capscrew Part Number 3093940.
 - Tighten capscrews to 75 N-m [55 ft-lb]
- Reinstall fan bracket.
 - Reference Procedure 008-088 in Section 8.
- Install idler pulley Part Number 3681587 to idler pulley bracket Part Number 5575127. Use capscrews Part Number 3 3903112 and washers Part Number 3 3026269.
 - Tighten capscrew to 46 N-m [34 ft-lb]
- Install dust shield Part Number 3935013 to idler pulley by pressing on outside diameter of shield to prevent damage to dust shield.
 - Reference Procedure 008-030 in Section 8.
- Install fan drive belt, Part Number 3103848.
 - Reference Procedure 008-002 in Section 8.

Service Parts

Service parts are available. See Tables 1 and 2 below.

Table 1, Service Parts for Engines Equipped with FA1542 Option

Part Number	Quantity	Part Description
70760	1	Dowel Pin
3009330	1	Washer
3026269	2	Washer
3035803	3	Hexagon Nut
3036460	1	Washer (spring)
3093936	3	Capscrew
3093940	3	Capscrew
3103848	1	Belt
3104029	1	Belt Tensioner
3681191	1	Bracket, Belt Tensioner
3681390	1	Capscrew
3681587	2	Idler Pulley (flat)
3681715	1	Pilot Adapter for Idler Pulley
3902460	1	Capscrew
3903112	1	Capscrew
3914407	1	Capscrew
3935013	2	Dust (Debris) Shield
5575127	1	Idler Pulley Bracket Assembly (includes bottom two idler pulleys)

Table 2, Service Parts for Engines Currently Built with FA1657 Option		
Part Number	Quantity	Part Description
3026269	1	Washer
3035803	3	Hexagon Nut
3093940	3	Capscrew
3103848	1	Belt
3681587	1	Idler Pulley (Flat)
3903112	1	Capscrew
3935013	1	Dust (Debris) Shield
5575127	1	Idler Pully Bracket Assembly (includes bottom two idler pulleys)

Production Status

Implemented for production. See Table 2.

Table 3, Production Information		
ESN Last	Build Date ¹	Plant
80074552	Nov 2018	Jamestown Engine Plant
¹ Engine build date can be found on engine dataplate.		

Document History

Date	Details
2018-10-15	Module Created